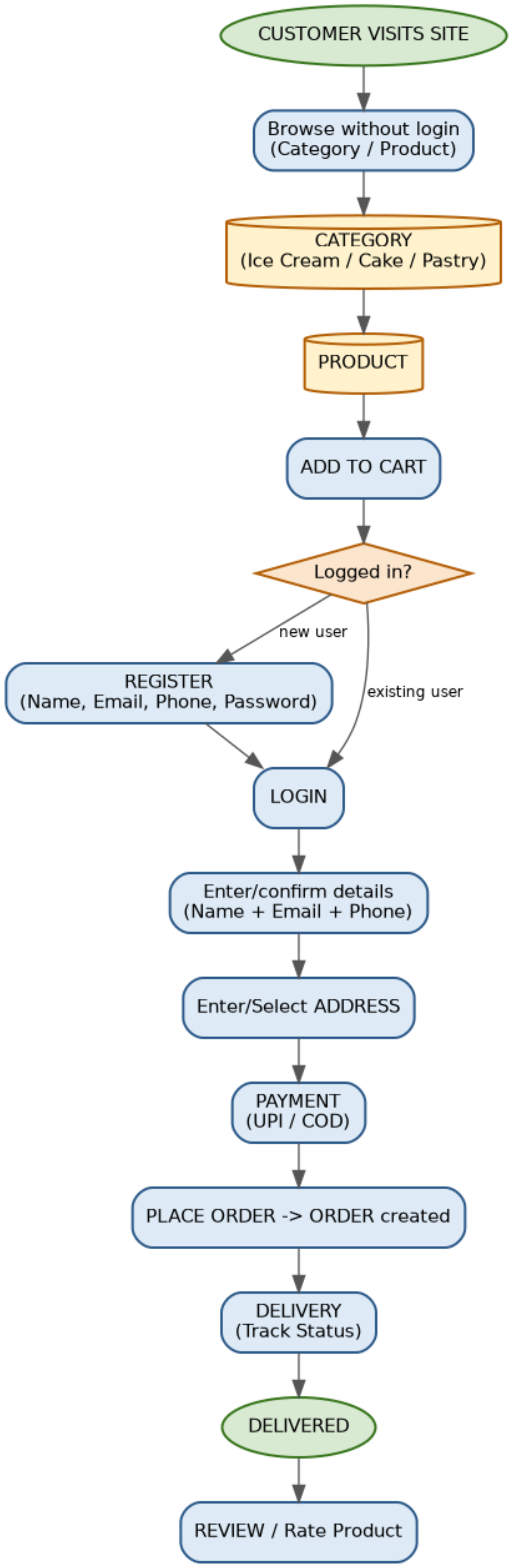
Online Bakery & Ice Cream Ordering System

System Flow & Entity-Relationship (ER) Diagram Documentation

Contents: Process Flow Diagram • Entity List (Strong / Weak) • Attributes with Primary & Foreign Keys • Relationship Summary • Full ER Diagram (Chen Notation)

1. System Process Flow

The diagram below traces the customer journey from browsing the catalogue (without needing to log in) through registration/login, cart, checkout, payment, delivery tracking, and finally leaving a review.



2. Entities Overview

The system is modelled using 5 **strong entities** (have their own primary key, exist independently) and 5 **weak entities** (cannot exist without an owner entity; identified using a partial key + the owner's primary key).

2.1 Strong Entities

Strong Entity	Description
CUSTOMER	A registered/browsing user of the site
CATEGORY	Product category: Ice Cream, Cake, Pastry
PRODUCT	An item that can be purchased
ORDERS	A confirmed order placed by a customer
PAYMENT	Payment made for an order (UPI / COD)

2.2 Weak Entities

Weak Entity	Owner Entity	Description
ADDRESS	CUSTOMER	A delivery address belonging to a customer
CART_ITEM	CUSTOMER	A product + quantity sitting in a customer's cart
ORDER_ITEM	ORDERS	A product + quantity that is part of a placed order
DELIVERY	ORDERS	Delivery/tracking record for an order
REVIEW	CUSTOMER	A rating/comment a customer leaves for a product

3. Attributes, Primary Keys & Foreign Keys

PK = Primary Key, FK = Foreign Key, PPK = Partial Key (discriminator of a weak entity).

CUSTOMER (strong)

Attribute	Key	Notes
customer_id	PK	Unique customer identifier
name		Customer's full name
email		Login email
phone		Contact number
password		Hashed login password

CATEGORY (strong)

Attribute	Key	Notes
category_id	PK	Unique category identifier
category_name		Ice Cream / Cake / Pastry

PRODUCT (strong)

Attribute	Key	Notes
product_id	PK	Unique product identifier
category_id	FK -> CATEGORY	Category the product belongs to
product_name		Name of the item
price		Unit price
description		Item description

ADDRESS (weak, owner: CUSTOMER)

Attribute	Key	Notes
address_id	PPK	Discriminator within a customer
customer_id	FK -> CUSTOMER	Owning customer
street / city / state / pincode		Address fields

CART_ITEM (weak, owner: CUSTOMER)

Attribute	Key	Notes
cart_item_id	PPK	Discriminator within a customer
customer_id	FK -> CUSTOMER	Owning customer
product_id	FK -> PRODUCT	Product added to cart
quantity		Number of units

ORDERS (strong)

Attribute	Key	Notes
order_id	PK	Unique order identifier
customer_id	FK -> CUSTOMER	Customer who placed the order
address_id	FK -> ADDRESS	Delivery address for this order
order_date		Date/time placed
total_amount		Order total
order_status		Placed / Confirmed / Cancelled etc.

ORDER_ITEM (weak, owner: ORDERS)

Attribute	Key	Notes
order_item_id	PPK	Discriminator within an order
order_id	FK -> ORDERS	Owning order
product_id	FK -> PRODUCT	Product ordered
quantity / price		Qty and price at time of order

PAYMENT (strong, 1:1 with ORDERS)

Attribute	Key	Notes
payment_id	PK	Unique payment identifier
order_id	FK -> ORDERS	Order being paid for
payment_mode		UPI / COD
payment_status		Pending / Success / Failed
amount / payment_date		Amount paid and when

DELIVERY (weak, owner: ORDERS)

Attribute	Key	Notes
delivery_id	PPK	Discriminator within an order
order_id	FK -> ORDERS	Order being delivered
delivery_status		Packed / Out for delivery / Delivered
delivery_date		Expected/actual delivery date

REVIEW (weak, owner: CUSTOMER)

Attribute	Key	Notes
review_id	PPK	Discriminator within a customer
customer_id	FK -> CUSTOMER	Reviewer
product_id	FK -> PRODUCT	Product being reviewed
rating / comment / review_date		Review content

4. Relationship Summary

Relationship	Entity 1	Cardinality	Entity 2	Identifying?
places	CUSTOMER	1 : M	ORDERS	No
has	CUSTOMER	1 : M	ADDRESS	Yes
adds	CUSTOMER	1 : M	CART_ITEM	Yes
writes	CUSTOMER	1 : M	REVIEW	Yes
belongs to	PRODUCT	M : 1	CATEGORY	No
refers to	CART_ITEM	M : 1	PRODUCT	No
contains	ORDERS	1 : M	ORDER_ITEM	Yes
refers to	ORDER_ITEM	M : 1	PRODUCT	No
is for	REVIEW	M : 1	PRODUCT	No
has	ORDERS	1 : 1	PAYMENT	No
has	ORDERS	1 : 1	DELIVERY	Yes
shipped to	ORDERS	M : 1	ADDRESS	No

Legend used in the ER diagram (next page):

Symbol	Meaning
Bold single-line rectangle (orange fill)	Strong entity
Double-line rectangle (yellow fill)	Weak entity
Single diamond (green)	Regular (non-identifying) relationship
Double diamond (blue)	Identifying relationship (connects owner to weak entity)
Ellipse, underlined, yellow	Primary key attribute
Ellipse, underlined dashed, pink	Partial key (discriminator) of a weak entity
Plain ellipse	Regular attribute

5. Full Entity-Relationship Diagram (Chen Notation)